Shubham Haraniya

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Education

- **M.Tech in Artificial Intelligence** – Indian Institute of Technology, Jodhpur *2025 – Present, CGPA: 9.0*
- **B.E. in Computer Engineering** – L.D. College of Engineering, Ahmedabad *2021 – 2025, CGPA: 8.49*

Experience

- **IntelKit Solutions** *Jan 2025 – June 2025*

Data Scientist Intern

  - Developed a **deepfake detection system** using computer vision and deep learning techniques to identify manipulated images and videos, implementing preprocessing, feature extraction, model training, and evaluation pipelines for reliable media authenticity analysis.
  - Designed and deployed automated data acquisition pipelines using Selenium and BeautifulSoup, enabling extraction and reducing manual data collection effort by **over 70%** for downstream NLP and Machine Learning workflows.

Projects

- **Deepfake Forensics – Frequency-Aware SE-ADDNet** *Github*

Deep Learning / Computer Vision Project Feb 2026 – April 2026

  - Proposed a novel **dual-stream architecture** jointly modeling spatial manipulation cues (EfficientNet-B5 + SE/ADD attention) and FFT-based spectral artifacts via a **learnable FrequencyGate fusion**, achieving **94.6% accuracy** and **96.3% F1** on 12K test images across 4 deepfake generators.
  - Served real-time **GradCAM forgery localization** via a Streamlit web app, providing interpretable region-level detection explanations.
  - **Technologies**: PyTorch, EfficientNet-B5 (timm), FocalLoss, FP16 Mixed Precision, GradCAM, Streamlit

- **EfficientUNet – Lightweight Semantic Segmentation** *Github*

Deep Learning / Computer Vision Project Feb 2026 – Mar 2026

  - Developed an ultra-lightweight **EfficientUNet** for 21-class segmentation on PASCAL VOC 2012, achieving **69.47% clean Dice** and **59.67% Dice** under stochastic sensor noise corruptions.
  - Compressed compute footprint to **0.079 GFLOPs** and **0.64M parameters** (∼4× reduction vs. baseline UNet) via **depthwise-separable convolutions**.
  - Implemented **PyTorch AMP** mixed-precision training, custom combined Dice + Cross-Entropy loss with boundary masking, TensorBoard tracking, and an automated **pytest** suite.
  - **Technologies**: PyTorch, Torchvision, Python, thop, TensorBoard, Pillow, pytest, Makefile

- **Multimodal Price Estimator** *Github*

Amazon ML Project Oct 2025 – Oct 2025

  - Developed an ensemble pricing model using **parallel MLPs** on **CoCa** and **SigLIP** multimodal embeddings, achieving an **sMAPE of 40.06%** on product price prediction.
  - Optimized the architecture for fast inference at **0.12s/sample** through efficient model design.
  - **Technologies**: PyTorch, CoCa, SigLIP, CLIP, Ensemble Learning

Technical Skills

- **Languages & Core:** Python, SQL, Data Structures & Algorithms, OOP
- **Data Science & ML:** NumPy, Pandas, SciPy, Scikit-learn, XGBoost, LightGBM, Random Forest, Clustering, PCA, Anomaly Detection, EDA, Feature Engineering, Statistical Analysis & Hypothesis Testing, Cross-Validation, Hyperparameter Tuning, Matplotlib, Seaborn
- **Deep Learning & Computer Vision:** PyTorch, TensorFlow, Keras, OpenCV, CNNs, YOLO, R-CNNs, U-Net, ViT, RNNs/LSTMs, Attention, Transfer Learning, Object Detection, Segmentation, Image Classification & Captioning
- **NLP & LLM Engineering:** Hugging Face Transformers, LLMs/SLMs/VLMs, Prompt Engineering, Pretraining, SFT, RLHF, DPO, PEFT (LoRA, QLoRA), TRL, Quantization
- **RAG & Agentic AI:** RAG (Hybrid Retrieval, Reranking, GraphRAG, Agentic RAG), Embeddings, Vector Databases, LangChain, LangGraph, ReAct, Tool Calling, Multi-Agent Systems, MCP, Context Engineering, Agent Memory
- **Evaluation, Safety & Explainability:** Precision/Recall/F1, ROC-AUC, RMSE, mAP, IoU, Error Analysis, SHAP, Grad-CAM, RAGAS, DeepEval, LangSmith, LLM Guardrails, Prompt Injection Defense, Tracing
- **MLOps & Deployment:** FastAPI, Docker, MLflow, Weights & Biases, vLLM, AWS, Git, PostgreSQL, REST APIs

Certifications

- **Python Data Structures and Algorithms** – Udemy *2025*
- **Machine Learning and Data Science** – Udemy *2023*

Achievements

- **Secured AIR 121**−in GATE Data Science and Artificial Intelligence 2025 with GATE Score of 780. *2025*
- **Secured Rank 11**−Amazon ML Challenge 2025 among 82,000+ teams across the country. *2025*